Listing 9.7 Circle pack code

```
function createPack(incData,targetSVG) {

   var depthScale = d3.scale.quantize()
         .domain([0,1,2]).range(colorbrewer.Reds[3]);

      packChart = d3.layout.pack();
   packChart.size([100,100])
        .children(function(d) {return d.values;})
        .value(function(d) {return 1;});

   d3.select(targetSVG)
        .append("g")
        .attr("transform", "translate(0,0)")
        .selectAll("circle")
        .data(packChart(incData))
        .enter()
        .append("circle")
        .attr("class", "pack")
        .style("fill", function(d) {return depthScale(d.depth);});

};
```

This could be anything; we'll redraw and recall the layout to do so in the redraw function.

9.1.5 Redraw: resizing based on screen size

The redraw function creates the visual elements of your charts based on the current size of the screen. Without the redraw function, your charts wouldn't display, because the circles in the circle pack have no xy information and the bars in your bar chart have no height or width. This may seem counterintuitive, but it's better to have your chart-drawing code in one place for easy maintainability. If you're going to constantly update a chart, as we're doing with our dashboard, then there's no reason at all to draw the elements (or set their graphical channels) when you first append them to the SVG canvas. By decoupling the preparation of the dataset from their drawing, you're doing the same thing that layout functions are designed to do in D3: not draw things, but process them for drawing. You can then call the redraw function, as shown in the following listing, not only when the screen resizes but whenever a change to filters or data warrants it. For example, we can filter the tweets to a certain period of time, which we'll do in section 9.3, "Brushing."

Listing 9.8 The `redraw` function

```
function redraw() {
    var leftSize = canvasSize("#leftSVG");
    packChart.size(leftSize)

    d3.select("#leftSVG")
       .selectAll("circle")
       .data(packChart(packableTweets))
       .attr("r", function(d) {return d.r;})
       .attr("cx", function(d) {return d.x;})
       .attr("cy", function(d) {return d.y;});
```

The pack layout creates new data based on the adjusted size of the SVG.